
School of Mathematics and Statistics

MT5864 Advanced Group Theory

Problem Sheet VI: Nilpotent Groups

1. Let G be a group and, for each integer $i \geq 1$, let $\gamma_i(G)$ denote the i -th term of the lower central series of G . Show that $\gamma_i(G)/\gamma_{i+1}(G) \leq Z(G/\gamma_{i+1}(G))$ for all i .
2. Let G and H be nilpotent groups. Show that the direct product $G \times H$ is nilpotent.
3. Calculate the terms in the lower central series for the following groups:

$$(i) S_3; \quad (ii) D_8; \quad (iii) D_{10}; \quad (iv) A_4; \quad (v) S_4.$$

4. Let G be a nilpotent group and N be a non-trivial normal subgroup of G . Take i to be the largest positive integer such that $\gamma_i(G) \cap N \neq 1$. Show that $[\gamma_i(G) \cap N, G] = 1$. Deduce that $Z(G) \cap N \neq 1$.
5. Let G be a group and define a collection of normal subgroups $Z_i(G)$ of G as follows: First set $Z_0(G) = 1$. Then, assuming that the normal subgroup $Z_i(G)$ has been defined for some integer $i \geq 0$, let $Z_{i+1}(G)$ be the normal subgroup of G such that

$$Z_{i+1}(G)/Z_i(G) = Z(G/Z_i(G)).$$

(That is, $Z_{i+1}(G)$ is the subgroup of G that corresponds to the centre of $G/Z_i(G)$ under the Correspondence Theorem.)

The chain

$$1 = Z_0(G) \leq Z_1(G) \leq Z_2(G) \leq \dots$$

is called the *upper central series* for G .

- (a) Which subgroup of G is equal to $Z_1(G)$?
- (b) Suppose that G is nilpotent and that $\gamma_{c+1}(G) = 1$. Show that $Z_i(G) \geq \gamma_{c+1-i}(G)$ for $i = 0, 1, \dots, c$. Deduce that $Z_c(G) = G$.
- (c) Suppose that $Z_c(G) = G$ for some c . Show that $\gamma_i(G) \leq Z_{c+1-i}(G)$ for $i = 1, 2, \dots, c+1$. Deduce that G is nilpotent.

In conclusion, this shows that, for any group G , $\gamma_{c+1}(G) = 1$ if and only if $Z_c(G) = G$ and hence an equivalent definition for a group to be nilpotent is that the upper central series eventually reaches G .

[Hint for (b) and (c): Use induction on i .]

6. Calculate the terms in the upper central series for the groups listed in Question 3.

7. A *maximal subgroup* of a group G is a proper subgroup M such that there is no subgroup H with $M < H < G$.

- (a) If G is a nilpotent group, show that every maximal subgroup is normal in G .
- (b) Deduce that every maximal subgroup of a nilpotent group has index p for some prime p .
- (c) Let G be a *finite* group and suppose that P is a Sylow p -subgroup of G which is not normal in G ; that is, $N_G(P) < G$. Use the Frattini Argument to show that if M is a maximal subgroup of G satisfying $N_G(P) \leq M < G$, then M is not normal in G .
- (d) Deduce that if every maximal subgroup of a finite group G is normal then G is nilpotent.

[Thus this gives another characterization of finite nilpotent groups: A finite group is nilpotent if and only if every maximal subgroup is normal.]

8. Let G be a finite group. The Frattini subgroup $\Phi(G)$ of G is the intersection of all its maximal subgroups.

- (a) Show that $\Phi(G)$ is a characteristic subgroup of G .
- (b) What is the Frattini subgroup of a finite simple group?
- (c) What is the Frattini subgroup of an elementary abelian p -group?
- (d) If X is a subset of G such that $G = \langle X, \Phi(G) \rangle$, show that $G = \langle X \rangle$.
[Hint: If $\langle X \rangle \neq G$, there exists a maximal subgroup that contains $\langle X \rangle$.]
- (e) Let P be a Sylow p -subgroup of the Frattini subgroup $\Phi(G)$. Use the Frattini Argument to show that P is a normal subgroup of G .
- (f) Deduce that the Frattini subgroup $\Phi(G)$ is always nilpotent.